



GLOBAL ECONOMICS
GRADE: 11TH
CATCH-UP
CRITERION C7
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SECOND
TERM
2025–2026

Learning objective: Construct and interpret a confidence interval for a population proportion.

Assessment criteria:
 C7: Constructs the confidence interval for a proportion.

Criteria assessment

Each assessment criterion is evaluated across the problems in this catch-up exam. A criterion is considered passed when it is correctly activated in 9 of the 10 problems of this activity.

1. Problem 1 — Problem description

A digital bank wants to estimate the proportion of young adult clients (ages 20–29) who use automatic monthly savings transfers. In a random sample of $n = 120$ clients, $x = 66$ report using automatic transfers. Construct and interpret a 90% confidence interval for the population proportion.

2. Problem 2 — Problem description

A rideshare platform wants to estimate the proportion of drivers in their 20s who bought private accident insurance this year. In a sample of $n = 200$ drivers, $x = 92$ purchased insurance. Construct 90%, 95%, and 99% confidence intervals for the population proportion, and give a brief interpretation.

3. Problem 3 — Problem description

A fintech app wants to estimate the proportion of users in their 20s who pay credit card balances in full each month. In a sample of $n = 260$ users, $x = 182$ report paying in full. Construct and interpret a 95% confidence interval. Then answer: is a benchmark value of $p = 0.75$ plausible based on this interval?

4. Problem 4 — Problem description

A micro-investing platform is evaluating a support program. It wants the proportion of young clients who invest every month to be at least 60%. After one cycle, a random sample of $n = 320$ clients finds $x = 173$ monthly investors. Construct a 98% confidence interval for the population proportion and decide whether the 60% benchmark is plausible. Justify your decision using the interval bounds.

5. Problem 5 — Problem description

A budgeting startup compares two groups of users in their 20s.

- Group A (users who completed a financial bootcamp): $n = 250$, $x = 170$ keep an emergency fund.
- Group B (users without bootcamp): $n = 250$, $x = 142$ keep an emergency fund.

Construct a 95% confidence interval for each group proportion. Compare the intervals and discuss what overlap or non-overlap suggests about relative plausibility of higher emergency-fund behavior.

6. Problem 6 — Problem description

A national survey studies young workers and compares two sectors.

- Sector 1 (gig logistics): $n = 300$, $x = 126$ contributed to retirement savings last month.
- Sector 2 (junior office jobs): $n = 280$, $x = 151$ contributed last month.

Construct 90% confidence intervals for both sector proportions. Which sector appears safer in terms of consistently higher retirement-saving participation, and which estimate is more uncertain?

7. Problem 7 — Problem description

A payment app tracks budgeting-rule usage (the 50-30-20 rule) among users in their 20s in two cities.

- City A: $n = 360$, $x = 198$ use the rule.
- City B: $n = 340$, $x = 170$ use the rule.

For each city, construct 90% and 95% confidence intervals. Then explain how the decision about whether City A is above 55% changes when the confidence level increases.

8. Problem 8 — Problem description

An online broker compares two periods for the same age group.

- Period 1: $n = 420$, $x = 231$ users made at least one ETF purchase.
- Period 2: $n = 410$, $x = 254$ users made at least one ETF purchase.

Construct a 95% confidence interval for each period. Compare uncertainty and discuss whether adoption appears to have increased.

9. Problem 9 — Problem description

A policy team evaluates three training channels that teach debt-management skills to young adults.

- Channel A: $n = 280$, $x = 168$ participants reduced debt for three consecutive months.
- Channel B: $n = 300$, $x = 171$ participants reduced debt for three consecutive months.
- Channel C: $n = 260$, $x = 130$ participants reduced debt for three consecutive months.

Construct 90% and 99% confidence intervals for all three channels. Then recommend which channel

should be prioritized if the target is to keep the true success proportion around or above 60%, considering uncertainty.

10. Problem 10 — Problem description

A youth-finance nonprofit pilots three versions of a savings habit program.

- Program X: $n = 350$, $x = 221$ participants stayed on a weekly savings plan for 12 weeks.
- Program Y: $n = 330$, $x = 191$ participants stayed on plan.
- Program Z: $n = 340$, $x = 180$ participants stayed on plan.

Construct 95% and 99% confidence intervals for each program. Then give a decision recommendation if funding should prioritize programs with a plausibly highest retention proportion and relatively stable uncertainty.